
MB 560 : Evolutionary computation for business solutions

Course Code : MB 560

Course Name : Evolutionary computation for business solutions

L-T-P-C : 2-0-0-2

Intended for : MBA

Prerequisite :

Mutual Exclusion : NA

Approval: 54th BoA

Course Contents

- **Introduction to business decision making and Optimization:** Overview of business decision making, Optimization Models for business problems, Traditional approaches: Linear and Non-Linear Methods. (3 Hours)
- **Overview of probability and sampling:** Overview of probability distribution, sampling, and random number simulation. (3 Hours)
- **Non-Traditional Methods:** Introduction to Evolutionary, Swarm and Nature inspired optimization techniques, Optimization approaches for single objective decision making problem, decision making under constraints: inequality and inequality constraint handling. (8 Hours)
- **Optimization under conflicting goals:** Introduction to Multi-objective optimization, Evolutionary Approaches for Multi-objective optimization: dominance, decomposition and preference based methods. (6 Hours)
- **Hybrid Approaches:** Hybrid approaches for solving problems, evolutionary and machine learning based wrapper approaches. (2 hours)
- **Evolutionary Computation application to business decision making:** Evolutionary Computation applications in: Project Management, Planning, Scheduling, Transportation, production and operations management, finance. (6 Hours)

Textbooks:

1. Biethahn, Jorg, and Volker Nissen, eds., **Evolutionary algorithms in management applications**, Springer Science & Business Media, 2012.
2. Burke, Edmund K., Edmund K. Burke, Graham Kendall, and Graham Kendall, **Search methodologies: introductory tutorials in optimization and decision support techniques**, Springer, 2014.
3. Relevant research articles and business cases

References:

1. NA